

CALRIG · REQUIREMENTS

CalRig requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/calrig		
Source	docs/03-requirements.md @ a22ff73		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	R1 redefined per CLR-DDR-001 A3; status column replaced by the CLR-CAL-001 results; budget note per A1
0.3	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with first-order status
0.1	2026-09-25	Amish Chadha	Initial scaffold

CalRig requirements

These requirements have been checked by calculation in CLR-CAL-001 v0.1. Targets are still proposals, not yet validated with users. R1 was redefined in this version under CLR-DDR-001 (item A3, adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review). Test conditions follow the enhanced lab conditions in the US EPA sensor testing reports (20 °C and 40 °C; 40 % and 85 % RH) so that CalRig results can be read against a known method ([EPA FAQ](#)).

The **reference room** is an indoor lab at 22 °C and 50 % RH unless a requirement states otherwise.

Table 1. Requirements with targets and status from CLR-CAL-001 (Table 4 there gives the values and script tags).

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CLR-CAL-001)
R1	Chamber temperature range	10 to 40 °C at the sensor tray in a room at 15 to 25 °C; in rooms from 25 to 30 °C, the lowest reachable set point is stated in each calibration record	Heat balance with a thermoelectric model; later chamber log	Met: 8.9 °C reachable in a 25 °C room; 13.3 °C lowest in a 30 °C room
R2	Chamber humidity range	20 to 85 % RH at any temperature from 20 to 40 °C	Moisture balance; later chamber log	Not met at 20 °C and 85 % RH in rooms above about 19 °C: the inner Peltier sink condenses faster than the bubbler supplies water. Other corners reached in 26 min or less
R3	Stability at a set point	±0.3 °C and ±2 % RH over a 30 min hold	Controller simulation; later log	Met on an idealized model: ±0.02 °C, ±1.0 % RH
R4	Uniformity across the six bays	±0.3 °C and ±2 % RH between bays	Airflow estimate; later traverse with two references	At risk: ±0.18 °C at the EPA points with the 120 mm fan, ±0.42 °C at 10 °C; ±0.8 % RH
R5	Reference temperature uncertainty	0.2 °C or better (expanded, k = 2) from 10 to 40 °C	Uncertainty budget; ice-point check	At risk: 0.14 °C with the typical sensor tolerance, 0.24 °C if it is 0.2 °C; no calibration certificate
R6	Reference humidity uncertainty	2 % RH or better (expanded) from 20 to 85 % RH, checked at four salt fixed points	Uncertainty budget; salt fixed-point checks (Greenspan, 1977)	At risk: 2.0 % RH, no margin
R7	Particle test range	Zero below 2 µg/m³ PM2.5; controlled decay from 300 to 5 µg/m³ in 45 min or less	Decay calculation; later run	Met: 5 min clean-down; 28 min decay
R8	Particle reference traceability	Transfer PM sensor collocated with a regulatory or research monitor for 30 days or more, and meeting EPA PM2.5 targets there, before use as the chamber reference	Collocation record	Not met by the rig alone. The collocation site is still open (CLR-DDR-001 O1)
R9	CO2 check	Zero check below 50 ppm with soda lime; span 400 to 2000 ppm against a reference of ±(30 ppm + 3 %)	Calculation and datasheet (Sensirion SCD30); later run	Met: 8 ppm floor; about 1,100 ppm added per liter of breath
R10	Capacity	Six sensor heads per run, each up to 90 x 70 x 50 mm, with power and I2C, UART or USB leads through a sealed gland	Model check	Met: six bays, 244 mm clear above the tray
R11	Throughput	Four-point temperature and humidity sweep plus one particle decay run in 8 h or less, unattended	Timing estimate; later log	Met: about 5.6 h
R12	Cost		Priced BOM (bom/ bom.csv)	Not met: \$404, over both \$300 and the proposed \$400

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CLR-CAL-001)
		\$300 or less in parts (budget_usd ; a rise to \$400 is proposed, awaiting Amish)		
R13	Size and mass	Fits a 600 x 500 mm bench area, 400 mm high or less; 12 kg or less	Model; mass estimate	Not met on mass: 600 x 500 x 374 mm fits; about 13.4 kg
R14	Records	Every run writes CSV (time, set points, references, each sensor) and a per-sensor report with fitted slope, offset and humidity term, EPA metrics and the reference check dates	Software review	Not verifiable at TRL 3 (software not written)
R15	Electrical safety	No mains wiring in the rig; certified external 12 V supply; fused 12 V bus; peak draw 100 W or less	Design review; power budget	Met: about 89 W, 7.4 A peak
R16	Thermal safety	Independent hardware cut-off opens the heater circuit at 50 °C air temperature or 70 °C Peltier hot side	Design review; heat balance	Met: outer sink 50 °C at most in normal use
R17	Aerosol and gas safety	No toxic gases; test smoke vented through the HEPA loop before the door opens; smoke source outside the chamber	Design review	Met by design
R18	Garage-buildable	Built with a saw or laser cutter, a drill and a soldering iron; all active parts off the shelf	Design review	Met by design

■ Requirements not met or at risk

- **R2**, humidity at 20 °C and 85 % RH. The inner Peltier sink runs about 1.2 K below the dew point in the reference room (CLR-CAL-001, section C). A larger inner sink is proposed in docs/REVIEW.md .
- **R8**, particle traceability. The chamber cannot make a mass reference; it transfers one from a collocation site that is still to be chosen.
- **R12**, cost. \$404 against \$300; the proposed \$400 is also exceeded.
- **R13**, mass. About 13.4 kg against 12 kg; the footprint and height now fit.
- **R4**, **R5** and **R6** are at risk (uniformity at cold points, sensor tolerance and humidity budget without margin).

■ Assumptions

- Chamber conductance about 0.82 W/K with the full jacket and door panel fitted, and 6.7 W of internal gains (CLR-CAL-001).
- Settling is limited by the 3 L/min conditioning flow, moisture held on the walls and a 15 min equilibration allowance for the heads.
- Test aerosol is incense smoke (CLR-DDR-001 A4); results for other aerosols (dust, sea salt) will differ and must be stated in each report.